Kinetic Theory of Non-Equilibrium Systems

Esther Esteban Huerga,[1] Javier Herrero Pérez,[1] and Irene Rodríguez Sánchez[1]
Student, Faculty of Physics, University of Salamanca

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Using kinetic theory, we study the transition from non-equilibrium to equilibrium in homogeneous systems, without external forces and in the relaxation time approximation. We employ the Boltzmann and Enskog-Boltzmann equations, together with Boltzmann's H-theorem, to analyze the dynamics of momentum distributions and the Boltzmann H-function. The study compares two initial conditions of velocities, normalized and non-normalized, using Monte Carlo simulations. Our results show the convergence of the system to equilibrium.

Keywords: Boltzman's H theorem, relaxation time approximation, homogeneous systems, forces free systems, average speed values, evolution to equilibrium, non-equilibrium systems, Monte Carlo simulation, kinetic theory, elastic spheres.

I. INTRODUCTION

The kinetic theory used in physics and chemistry explains the macroscopic behavior and properties of gases from a statistical study of microscopic molecular processes.

Using this theory, we intend to study the evolution of an out-of-equilibrium system, which evolves towards equilibrium, formed by a set of particles (N), which collide with each other. For this a statistical description is useful, by means of the introduction of the distribution function $f(\mathbf{r},\mathbf{v},t)$. In this way, it is possible to study the evolution of the velocity as the system evolves towards equilibrium, using averaged quantities.

Often in kinetic theory, the Maxwell-Boltzmann distribution is used as the velocity distribution describing the system at equilibrium in the Equation (1).

$$f(\mathbf{v}) = n\left(\frac{m}{2\pi kT}\right)^{3/2}\exp\left[-\frac{m(\mathbf{v}-\mathbf{u})^2}{2kT}\right] \tag{1}$$

However, systems do not have to start from equilibrium. If we have a system out of equilibrium, it is useful to use the Equation (2) developed by Ludwig Boltzmann, which is applied in homogeneous systems without external forces, and describe the time evolution of the velocity distribution function.

$$\left(\frac{\partial f}{\partial t}\right) = \int d^3\mathbf{v}_1 \int d\Omega_{cm}\sigma_{cm}(\Omega_{cm})|\mathbf{v}_1-\mathbf{v}|(f_1'f'-ff_1) \tag{2}$$

Normally, the aim is to study the evolution of the velocities of the system, then the averaged distribution function is $\langle \mathbf{v}^p \rangle$.

In these cases, the system is studied with simulations, since the Boltzmann equation cannot be solved theoretically for averages, so an approximation is used.

With this we introduce the approximation of the Boltzmann relaxation time Equation (3) which we will use later for the demonstration of the Equation (9) in the next Sec. II.

$$\left(\frac{\partial f}{\partial t}\right) + \mathbf{v}\boldsymbol{\nabla}_r f + \frac{\mathbf{F}}{m}\boldsymbol{\nabla}_v f = \left(\frac{\partial f}{\partial t}\right)_{collisions} = -\frac{f-f^{(0)}}{\tau_0} \tag{3}$$

where $f^{(0)}$ is the local equilibrium distribution function and τ_0 we introduce as the relaxation time.

As we are working on an homogeneous and without external forces system, Equation (3) simplifies to:

$$\left(\frac{\partial f}{\partial t}\right) = -\frac{f-f^{(0)}}{\tau_0} \tag{4}$$

If we evaluate the $\left(\frac{\partial f}{\partial t}\right)_{collisions}$ part, we can put the Equation (4) as the Boltzmann Equation (2).

However, not only the distribution of velocities is studied, but also the H-function evolution. This theorem tells us that if the time evolution can be described by means of the Boltzmann equation, then the H-function tends to decrease. Where the H function has been defined as follows in the Equation (5).

$$H(t) \equiv \int d^3\mathbf{r} \int d^3\mathbf{v} f(\mathbf{r},\mathbf{v},t) \ln f(\mathbf{r},\mathbf{v},t) \tag{5}$$

Returning to the evolution of the system, as we have already mentioned, the Boltzmann equation is able to describe an out-of-equilibrium collision system and to give its evolution in time. Let us now introduce the Enskog-Boltzmann Equation (6), by using the rules from the article[1], we arrive to a expression for elastic hard spheres[2].

$$\partial_t f(\mathbf{v}_1,t) = g_2(\boldsymbol{\sigma}_{12})\boldsymbol{\sigma}_{12}^{d-1} \int d\mathbf{v}_2 \int_{\mathbf{v}_{12}\cdot\hat{\boldsymbol{\sigma}}_{12}>0} d\hat{\boldsymbol{\sigma}}_{12} (\mathbf{v}_{12}\cdot\hat{\boldsymbol{\sigma}}_{12}) \times \{f(\mathbf{v}_1^{**},t)f(\mathbf{v}_2^{**},t) - f(\mathbf{v}_1,t)f(\mathbf{v}_2,t)\} = g_2(\boldsymbol{\sigma}_{12})I(f,f) \tag{6}$$

where d the dimension of the system, and $g_2(\boldsymbol{\sigma}_{12})$ is the pair correlation function of hard spheres at contact given by Enskog. This equation turns out to be a modification of the Boltzmann Equation (3) for hard spheres that takes into account the particle size through the $g_2(\boldsymbol{\sigma}_{12})$ term in this equation.

This paper examines the evolution of a system from a non-equilibrium state to equilibrium using kinetic theory. We focus on a homogeneous, isolated system undergoing elastic collisions.

Our main objectives will focus on studying the evolution of systems from non-equilibrium to equilibrium within the relaxation-time approximation. To this end, we aim to obtain the value of the relaxation time τ_0 and to demonstrate that the equilibrium state is independent of the initial conditions. We will observe the differences in relaxation times and their equilibrium values for two distinct initial velocity conditions. Additionally, we will examine how the system changes when varying the number of particles and how the evolution occurs for different distributions of $\langle \mathbf{v}^p \rangle$.

We simulate the system's evolution toward equilibrium using a Monte Carlo method. The velocity distribution is measured during the simulation to assess the system's progress toward equilibrium. Two initial velocity conditions will be compared: a spherical (normalized) and a cubic (non-normalized) condition.

The equilibrium state is approximated by the Maxwell-Boltzmann (MB) distribution. The theoretical framework includes the MB distribution, which the system is assumed to approach to at equilibrium.

Other approximations will also be necessary, such as the relaxation time. We will use this to describe the evolution of the velocity distribution and the H-function as a function of time.

Finally, the paper is divided into four main sections. The theory section (Sec. II), where theoretical developments are seen to introduce the Enskog-Boltzmann equation and the time evolution of the H-function in a more rigorous way; and the expression of the H-function and the velocity distribution averaged as a function of p. But it also introduces the relevant approaches that will be used in the simulation. The methodology section (Sec. III), which focuses on explaining the steps followed in the development of the simulation. The results (Sec. IV), which shows the graphs, tables and values obtained from the simulation, comparing them with the theoretical values of the theoretical analysis (introduced in theory). And finally, the conclusions (Sec. V) summarize the work, where we have started from and how far we have come, using the discussion of the results.

II. THEORETICAL PREDICTIONS

As previously stated, we are working with a homogeneous system, without external forces and with elastic collisions. With this in mind, we proceed to demonstrate the Enskog-Boltzmann Equation (6).

Our goal is to analyze the conservation of a quantity ψ, within velocity space as influenced by collisions.

We start with Equation (6), noting that the term $g_2(\boldsymbol{\sigma}_{12})$ represents the particle size.

Due to the fact that we want to see how the system evolves, we analyze the rate of change of the average $\langle \psi \rangle = (1/n) \int d\mathbf{v} \psi(\mathbf{v}) f(\mathbf{v},t)$, where $n = N/V$.

If we differentiate with respect to time and use the Equation (6), we reach the next one

$$\frac{d\langle\psi\rangle}{dt} = \frac{1}{n}\int d\mathbf{v}\psi(\mathbf{v})\partial_t f(\mathbf{v},t) = \frac{1}{n}\int d\mathbf{v}\psi(\mathbf{v})g_2(\boldsymbol{\sigma}_{12})I(f,f) \tag{7}$$

If we use symmetry in the variables $1 \leftrightarrow 2$ and define $\Delta\psi(v_i) = \psi(\mathbf{v}_i') - \psi(\mathbf{v}_i)$, then we can write de Equation (9).

$$\frac{d\langle\psi\rangle}{dt} = \frac{g_2(\boldsymbol{\sigma}_{12})\sigma_{12}^{d-1}}{n}\int d\mathbf{v}_1\, d\mathbf{v}_2 \int' d\hat{\boldsymbol{\sigma}}_{12}(\mathbf{v}_{12}\cdot\hat{\boldsymbol{\sigma}}_{12})f(\mathbf{v}_1,t)f(\mathbf{v}_2,t)\frac{\Delta[\psi(\mathbf{v}_1)+\psi(\mathbf{v}_2)]}{2} \tag{8}$$

Given that we are working with a velocity distribution, we now aim to derive the general expression to study this evolution towards equilibrium in our system, using the relaxation time approximation. We will then specialize it for the specific cases we are working with.

Now we want to consider a homogeneous system with no external forces, and analyze the evolution of mean values. For this, we use a general development that after we particularize for each case. We define the medium value as it follows

$$\langle\mathbf{v}^p\rangle = \frac{1}{n}\int d^3\mathbf{v}\cdot\mathbf{v}^p f(\mathbf{v},t) \tag{9}$$

Using the approximation of the relaxation times (Equation (4)) we have

$$\partial_t\langle\mathbf{v}^p\rangle = \frac{1}{n}\int d^3\mathbf{v}\cdot\mathbf{v}^p\partial_t f(\mathbf{v},t) = -\frac{1}{n}\int d^3\mathbf{v}\cdot\mathbf{v}^p\frac{f-f^{(0)}}{\tau_0} = -\frac{1}{n\tau_0}\left[\int d^3\mathbf{v}\cdot\mathbf{v}^p f - \int d^3\mathbf{v}\cdot\mathbf{v}^p f^{(0)}\right] = -\frac{1}{\tau_0}\left[\langle\mathbf{v}^p\rangle - \langle\mathbf{v}^p\rangle_0\right] \tag{10}$$

where $\langle\mathbf{v}^p\rangle_0$ is the medium value of the MB distribution. Solving the integral we arrive to

$$\langle\mathbf{v}^p\rangle(t) = \langle\mathbf{v}^p\rangle_0 + \left(\langle\mathbf{v}^p\rangle|_{(t=0)} - \langle\mathbf{v}^p\rangle_0\right)e^{-t/\tau_0} \tag{11}$$

We have to determine the medium value of the MB distribution $\langle\mathbf{v}^p\rangle_0 = \int d^3\mathbf{v}\cdot\mathbf{v}^p f^{(0)}$ with $f^{(0)} = \frac{A}{n}e^{\frac{-m\mathbf{v}^2}{2kT}}$ where $A = n\left(\frac{m}{2\pi kT}\right)^{3/2}$.

Finally we get to the most general expression for the time evolution.

$$\langle\mathbf{v}^p\rangle(t) = \frac{2^{1+\frac{p}{2}}\left(\frac{kT}{m}\right)^{p/2}\Gamma\left(\frac{3+p}{2}\right)}{\sqrt{\pi}} + \left(\langle\mathbf{v}^p\rangle|_{(t=0)} - \frac{2^{1+\frac{p}{2}}\left(\frac{kT}{m}\right)^{p/2}\Gamma\left(\frac{3+p}{2}\right)}{\sqrt{\pi}}\right)e^{-t/\tau_0} \tag{12}$$

Now we can particularize each case. For the mean value of the squared velocity $\langle\mathbf{v}^2\rangle$, its evolution respect time is zero. This is because we are considering that the energy of the system is conserved in the collisions, $\partial_t\langle\mathbf{v}^2\rangle = 0$. And its value corresponds to the average value of the distribution at equilibrium $\langle\mathbf{v}^2\rangle = 3k_BT/m$.

Let's now move on to define the H-function evolution. We will employ the relaxation-time approximation as well to express this evolution as a function of this time.

Following this, we determine Boltzmann's H-function using the Boltzmann equation within the relaxation-time approximation for a force-free and homogeneous system. Starting from Equation (5), we can observe its evolution with respect to time[3].

$$\frac{dH(t)}{dt} = \int d^3\mathbf{r}\int d^3\mathbf{v}\left(\frac{\partial f}{\partial t}\ln f + \frac{\partial f}{\partial t}\right) \tag{13}$$

where we must differentiate between $\partial_t f(\mathbf{v},t)_{collision}$ and $\partial_t f(\mathbf{v},t)_{flow}$, because the last one is zero due to Gauss's theorem, since it is an isolated system. And for the same reason we have that $\int d^3\mathbf{r} \int d^3\mathbf{v} \partial_t f = \partial_t \left[\int d^3\mathbf{r} \int d^3\mathbf{v} f\right] = \partial_t N = 0$. So using the relaxation time approximation Equation (4) we reach to

$$\frac{dH(t)}{dt} = -\frac{1}{\tau_0}\left[\underbrace{\int d^3\mathbf{r}\int d^3\mathbf{v} f \ln f}_{H} - \int d^3\mathbf{r}\int d^3\mathbf{v} f^{(0)} \ln f\right] \tag{14}$$

We additionally perform another approximation considering $f^{(0)} \ln f \approx f^{(0)} \ln f^{(0)}$, so that the second term corresponds to H-function in equilibrium. And solving the integral we arrive to the Equation (15), where we can write $H_0(t) = \int d^3\mathbf{r} \int d^3\mathbf{v} f^{(0)} \ln f \sim \int d^3\mathbf{r} \int d^3\mathbf{v} f^{(0)} \ln f^{(0)}$

$$H(t) = H_0 + (H(t=0) - H_0)e^{-t/\tau_0} \tag{15}$$

Finally, we need to calculate the equilibrium value H_0, using the MB distribution function, so in Equation (5) we substitute $f^{(0)} = \frac{A}{n} e^{\frac{-m\mathbf{v}^2}{2kT}}$

$$H_0 = A\int d^3\mathbf{r}\int d^3\mathbf{v} e^{-\frac{m\mathbf{v}^2}{2kT}} \ln\left(Ae^{-\frac{m\mathbf{v}^2}{2kT}}\right) == A\left\{\int d^3\mathbf{r}\int d^3\mathbf{v} e^{-\frac{m\mathbf{v}^2}{2kT}} \ln(A) - \int d^3\mathbf{r}\int d^3\mathbf{v} e^{-\frac{m\mathbf{v}^2}{2kT}}\left(-\frac{m\mathbf{v}^2}{2kT}\right)\right\} \tag{16}$$

where solving the spatial integral gives the Equation (17).

$$H_0 = A\ln A \cdot V \cdot 4\pi \int_0^\infty v^2 e^{-\frac{m\mathbf{v}^2}{2kT}} d\mathbf{v} - A \cdot \frac{m}{2kT} \cdot V \cdot 4\pi \int_0^\infty \mathbf{v}^4 e^{-\frac{mv^2}{2kT}} d\mathbf{v} \tag{17}$$

Solving the integral we arrive to the Equation (18).

$$H_0 = AV \cdot 4\pi\left[\ln A \cdot \frac{\sqrt{\pi}}{4}\left(\frac{2kT}{m}\right)^{3/2} - \frac{m}{2kT} \cdot \frac{3\sqrt{\pi}}{8}\left(\frac{2kT}{m}\right)^{5/2}\right] = AV\pi\left(\frac{2kT}{m}\right)^{3/2}\left[\ln A - \frac{3}{2}\right] \tag{18}$$

In the end, we include the A value as $AV\left(\frac{2\pi kT}{m}\right)^{3/2} = N$, so we can replace like $\ln A = \ln\left(\frac{V}{N}\right) + \frac{2}{3}\ln\left(\frac{2\pi kT}{m}\right)$. The final value using the MB distribution for the H-fucntion is

$$H_0 = N\left[\ln\left(\frac{V}{N}\right) + \frac{2}{3}\ln\left(\frac{2\pi kT}{m}\right) - \frac{2}{3}\right] \tag{19}$$

With this we can reach the solution of the H-function depending of the relaxation time.

$$H(t) = \left[N\ln\left(\frac{N}{V}\right) + \frac{3}{2}N\ln\left(\frac{m}{2\pi kT}\right) - \frac{3}{2}N\right]\left(1 - e^{-t/\tau_0}\right) + H(t=0)e^{-t/\tau_0} \tag{20}$$

III. METHODOLOGY

We will use the direct simulation Monte Carlo (DSMC) method, an efficient numerical tool for solving the Boltzmann-Enskog equation in nonequilibrium systems, as described in the work of Montanero and Santos[1]. This method is suitable for studying particle dynamics in homogeneous systems without external forces.

In this study, DSMC is employed to simulate a system of elastic particles in three dimensions, under initial conditions far from equilibrium. The method represents the velocity distribution through a set of simulated particles, whose velocities evolve through two main stages: collisions and free flow. The particles move freely between collisions, updating their velocities according to the external forces applied (in this case, none, since the system is homogeneous and without external forces).

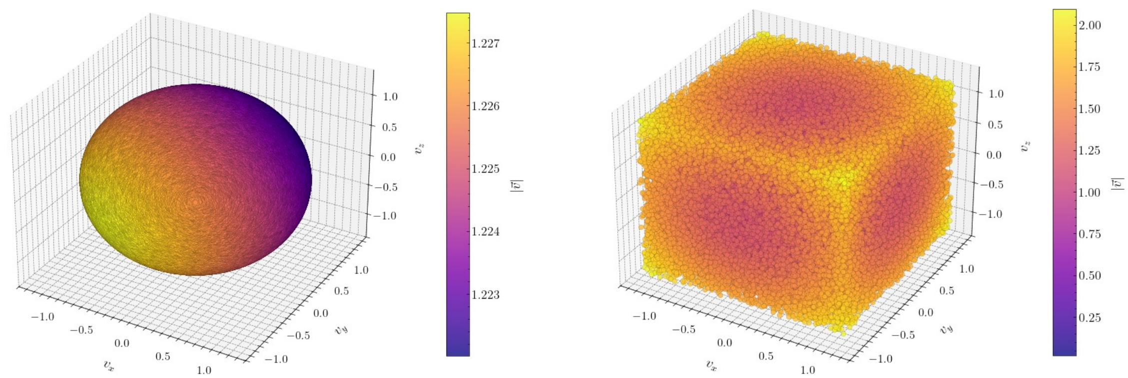
We develop a computational model to study relaxation processes in non-equilibrium systems approaching Maxwell-Boltzmann equilibrium. At equilibrium, the velocity distribution follows the Maxwell-Boltzmann (MB) distribution.

$$f(|\mathbf{v}|) = 4\pi \mathbf{v}^2 \pi^{-3/2} e^{-\mathbf{v}^2} \tag{21}$$

where the term $4\pi\mathbf{v}^2$ arises from the transformation to spherical coordinates.

The time in the simulation corresponds to the number of Monte Carlo (MC) steps. Each MC step represents one collision attempt between particles, and the physical time is related to the number of collisions per particle.

Two types of initial velocity distributions are used to generate the starting conditions for the simulation (Figure 1).



(a) Random (Unit Sphere)

(b) Random (Cube)

FIG. 1: Different initial velocity conditions, both with uniform distribution. Figure 1a has a module of 1, while Figure 1b is not normalized.

For the first initial condition “Random (unit sphere)” (Figure 1a), the particle velocities are generated using spherical coordinates of radius 1. the particle velocities are generated using spherical coordinates of radius 1. Two random values U_1 and U_2 are defined, where:

$$\theta \to \arccos(2U_1 - 1) \quad \phi \to 2\pi U_2 \tag{22}$$

Therefore, the velocities:

$$v_x = \sin\theta\cos\phi \quad v_y = \sin\theta\sin\phi \quad v_z = \cos\theta \tag{23}$$

The second initial condition, called “Random (Cube)” (Figure 1b), generates each velocity component independently and uniformly within the interval $[-1, 1]$, without imposing any constraint on the overall magnitude.

Both methods feature a uniform velocity distribution.

Before time evolution begins, we need to ensure that the total momentum of the system is zero, the velocity of the center of mass is first computed as:

$$\mathbf{v}_{\text{CM}} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{v}_i \tag{24}$$

where N is the total number of particles. Each particle’s velocity is then adjusted by subtracting the center of mass velocity:

$$\mathbf{v}_i \to \mathbf{v}_i - \mathbf{v}_{\text{CM}} \tag{25}$$

This operation guarantees that the net momentum of the system is zero, which is essential for isolating thermal effects in the simulation.

After this correction, the velocities are rescaled to match a target temperature T. This is done by applying a scaling factor λ, defined as:

$$\lambda = \sqrt{\frac{3NT}{\sum_i \left(v_{xi}^2 + v_{yi}^2 + v_{zi}^2\right)}} \tag{26}$$

which ensures that the average kinetic energy per particle corresponds to the desired temperature. The velocities are then updated according to:

$$\mathbf{v}_i \to \lambda \cdot \mathbf{v}_i. \tag{27}$$

This normalization step is crucial to start the simulation in a controlled thermal state, especially when studying equilibration dynamics or temperature-dependent properties.

The collision dynamics in the simulation develop according to different steps for each collision attempt as shown in Figure (2).

A random unit vector $\hat{\sigma}$ is generated, representing the direction of the collision. Then two distinct particles, labeled p_1 and p_2, are randomly selected from the system.

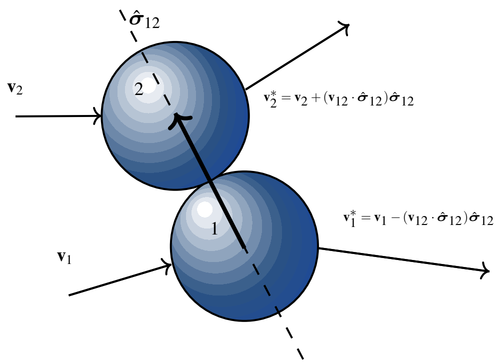


FIG. 2: Schematic of a collision between particles with initial velocities $\mathbf{v}_1$, $\mathbf{v}_2$ and post-collision $\mathbf{v}_1^*$, $\mathbf{v}_2^*$, in the contact direction $\hat{\sigma}_{12}$.

The relative velocity between the two particles is computed as:

$$\mathbf{v}_{12} = \mathbf{v}_1 - \mathbf{v}_2. \tag{28}$$

Next, the scalar product between the relative velocity and the collision normal is evaluated:

$$\omega = \mathbf{v}_{12} \cdot \hat{\sigma}_{12}. \tag{29}$$

The collision is accepted with a probability proportional to $|\omega|/\omega_{\max}$, where $\omega_{\max}$ is the current maximum value of $|\omega|$, updated dynamically during the simulation to maintain proper normalization.

If the collision is accepted, the velocities of the two particles are updated to conserve both momentum and kinetic energy:

$$\begin{aligned} \mathbf{v}_1^* &= \mathbf{v}_1 - (\mathbf{v}_{12} \cdot \hat{\sigma}_{12})\hat{\sigma}_{12} \\ \mathbf{v}_2^* &= \mathbf{v}_2 + (\mathbf{v}_{12} \cdot \hat{\sigma}_{12})\hat{\sigma}_{12} \end{aligned} \tag{30}$$

This procedure replicates binary collisions, ensuring the macroscopic properties of the system evolve according to kinetic theory, while enabling thermalization toward equilibrium.

Velocity moments $\langle \mathbf{v}^p \rangle$ are computed by sampling every N collisions, optimizing computational efficiency since high-frequency sampling is unnecessary for accurate statistical averages.

The H-function is calculated using a histogram of particle speeds, constructed between the minimum and maximum observed values. The histogram is divided into fixed bins and normalized so that the total area equals one, allowing it to be treated as a probability density function for entropy-related evaluations.

Natural units are employed throughout the simulation, setting quantities such as mass, Boltzmann's constant, and characteristic velocity scales to unity, simplifying numerical implementation and result interpretation.

Finally, we specify the simulation parameters used in our study:

- Temperature: $T = 0.5$
- Number of particles: $N = 10^5$
- Total number of collisions: 200
- Histogram resolution for the H-function: 1000 bins

- The entire simulation is repeated 100 times, and the final results are averaged to reduce statistical fluctuations
- The simulation is carried out in natural units

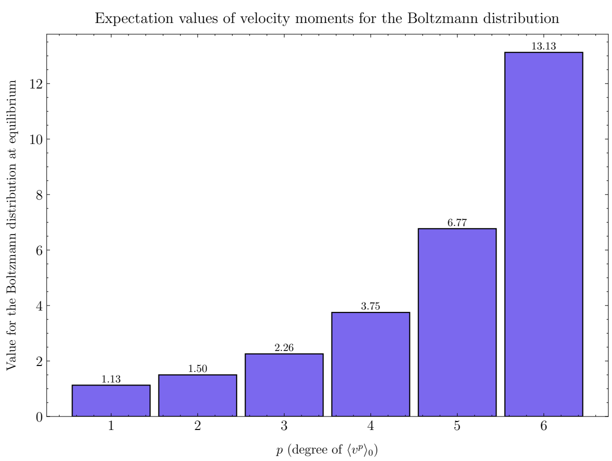
These configurations facilitate computational simulation and statistical accuracy, allowing an analysis of the system's approach to equilibrium and validation of the H-theorem.

This approach combines exact conservation laws with sampling techniques, enabling the study of the evolution of nonequilibrium phenomena while maintaining comparability with the predictions of kinetic theory. The construction of initial conditions, the precise application of conservation laws during time evolution, and the systematic approach to measuring relaxation processes provide a basis for studying how nonequilibrium systems approach thermal equilibrium.

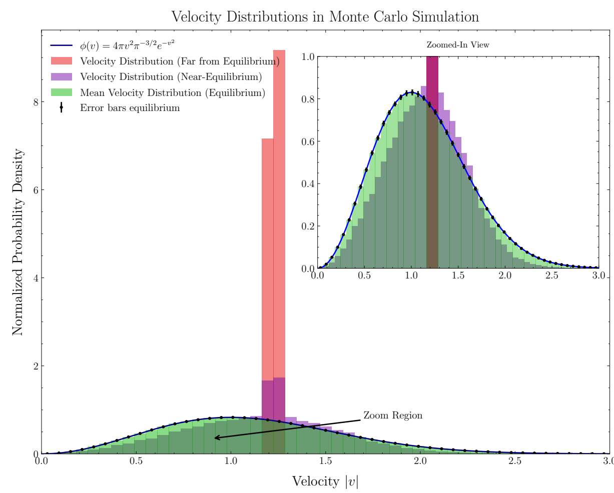
As we have said, the collisions take place in the normal direction, and its follows the rules given in the Montaneros article[1].

IV. RESULTS

Initially, we consider a random velocity distribution where the magnitude (or modulus) is set to 1. We will also study the system with another speed initial condition, starting from a random distribution without this normalization (magnitude not equal to 1) (Figure 1).



(a) Representation of the possible values of $\langle \mathbf{v}^p \rangle$ at equilibrium for a temperature of $T = 0.5$ in natural units for the theorical values.



(b) Representation of the MB distribution in the Monte Carlo Simulation, for three cases of envolve of the system: far from equilibrium, near, and in the equilibrium.

FIG. 3: Representation of the values of $\langle \mathbf{v}^p \rangle$ in Figure 3a. And of the MB distribution in the Monte Carlo simulation in Figure 3b.

At first, by using the Equation (12), the variation of $\langle \mathbf{v}^p \rangle$ is represented against that of p, in order to see the evolution of the medium value for a given temperature.

Figure 3a illustrates how the equilibrium value increases exponentially with p.

As the system evolves, the velocity distribution will tend towards a Maxwell-Boltzmann distribution, as shown in Figure 3b.

Now that we have examined the equilibrium velocity distribution of the system, we can begin to simulate the evolution of the $\langle \mathbf{v}^p \rangle$ velocity and compare it with its theoretical value.

The equilibrium value reached by the velocity distribution has been plotted for various values of p (Table I and Table II).

First, we will perform this simulation for the distribution with a norm equal to one, and then we will compare the results with the distribution that does not have a norm of one (Figure 4).

In this section, we will use equation (12). The case where $p = 2$ is straightforward because $\langle \mathbf{v}^2 \rangle$ remains constant over time, due to the conservation of energy that we have imposed.

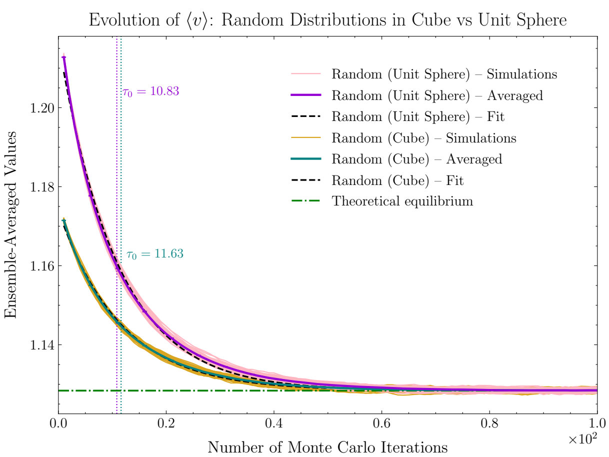
Here, we observe two distinct evolutionary behaviors: that of $\langle \mathbf{v} \rangle$ and those for $p > 2$. The evolution of $\langle \mathbf{v} \rangle$ shows a decrease, while the evolutions for $p > 2$ exhibit an increase with the number of Monte Carlo iterations. This behavior is attributed to the specific type of distribution we have chosen.

Random (Unit Sphere)			
p	$\tau_0 \pm \delta\tau_0$	$\langle v^p \rangle_{t=0} \pm \delta\langle v^p \rangle$	R^2
1	10.83 ± 0.06	1.2168 ± 0.0004	0.9983
3	11.61 ± 0.04	1.861 ± 0.001	0.9993
4	12.33 ± 0.03	2.310 ± 0.002	0.9997
5	13.19 ± 0.01	2.824 ± 0.002	0.9999
6	14.09 ± 0.02	3.28 ± 0.01	0.9998

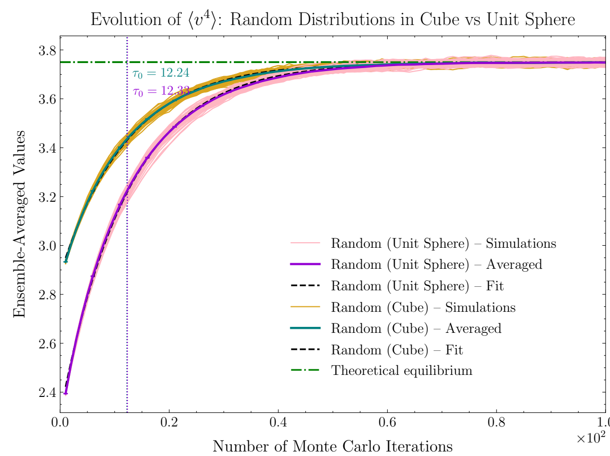
TABLE I: Fitted values of Random (Unit Sphere) for different dimensions.

Random (Cube)			
p	$\tau_0 \pm \delta\tau_0$	$\langle v^p \rangle_{t=0} \pm \delta\langle v^p \rangle$	R^2
1	11.63 ± 0.04	1.1739 ± 0.0001	0.9992
3	11.94 ± 0.04	2.0335 ± 0.005	0.9994
4	12.24 ± 0.03	2.881 ± 0.002	0.9996
5	12.69 ± 0.02	4.216 ± 0.003	0.9998
6	13.21 ± 0.01	6.311 ± 0.005	0.9999

TABLE II: Fitted values of Random (Cube) for different dimensions.



(a) Figure of the distributions for $\langle \mathbf{v} \rangle$.



(b) Figure of the distributions for $\langle \mathbf{v}^4 \rangle$.

FIG. 4: The Figure 4a correspond to the evolution of $\langle \mathbf{v} \rangle$ with the number of Monte Carlo iterations; and the 4b for the $\langle \mathbf{v}^4 \rangle$ distribution of the velocities .

Another important observation is the varying quality of the fit between the theoretical value and the evolution of $\langle \mathbf{v}^p \rangle$ for the several values of p considered (for more information regarding the figures for other values of p, please refer to Appendix B). As p increases, the fit with the theoretical value becomes poorer. This is because working with larger powers amplifies the error.

To observe how the velocity distribution develops with variations in the number of particles in the system (N), we have plotted $\langle \mathbf{v}^4 \rangle$ for different values of N (Figure 5a).

We observe a difference in the evolution based on the number of particles (N). For higher values of N, the system reaches equilibrium in fewer iterations.

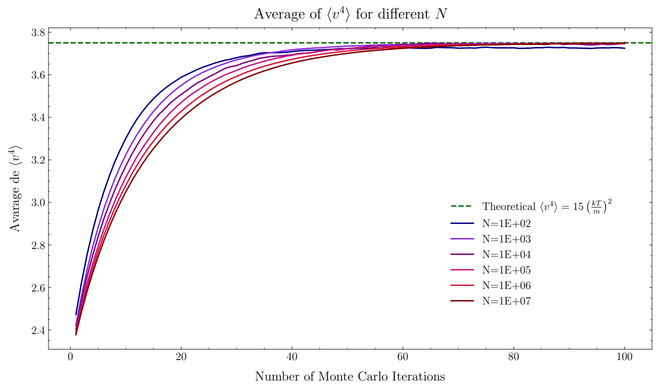
A clarification is necessary: for smaller N, the system did not reach equilibrium as precisely as it did for larger N. This is because we are simulating a finite-sized system intended to represent an infinite stochastic system.

We have plotted the relaxation time for different values of N for the case of $\langle \mathbf{v}^4 \rangle$, allowing us to visualize the system's evolution and its associated errors (Figure 5b).

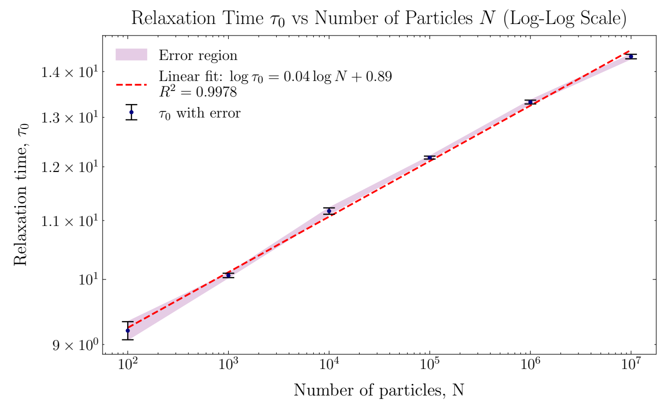
Distribution	$\boldsymbol{\tau_0 \pm \delta\tau_0}$	$\boldsymbol{H_{(t=0)} \pm \delta H_{(t=0)}}$	$\boldsymbol{R^2}$
Cube	4.72 ± 0.05	-3.086 ± 0.001	0.9956
Unit Sphere	3.57 ± 0.02	0.98 ± 0.02	0.9998

TABLE III: Relaxation time and H-function at equilibrium for different initial distributions.

Therefore, we determined that the larger N is, the longer it takes for the system to reach equilibrium, as we had previously noted.

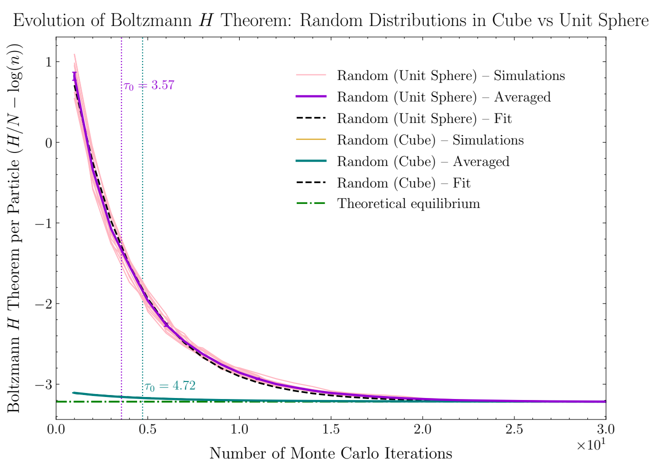


(a) Evolution of the $\langle \mathbf{v}^4 \rangle$ for different values of N.

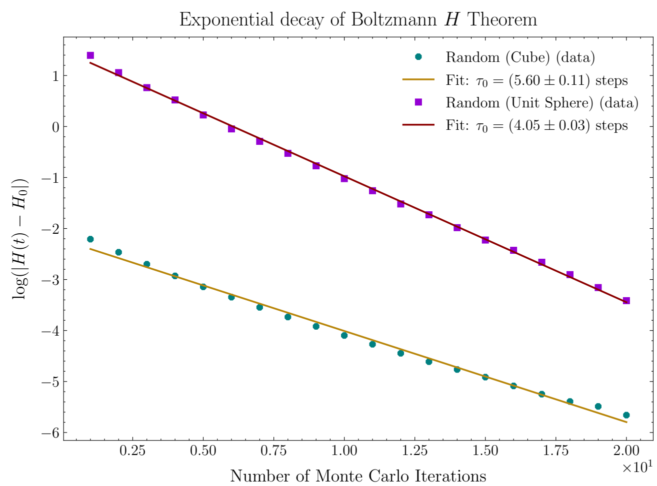


(b) Evolution of τ_0 for different values of N in logarithmic scale, for the case of $\langle \mathbf{v}^4 \rangle$.

FIG. 5: Representation of the evolution of $\langle \mathbf{v}^4 \rangle$ for different values of N for Figure 5a, and the evolution of τ_0 for different values of N in logarithmic scale for Figure 5b.



(a) Evolution of the H-function.



(b) Evolution in logarithmic scale of the H-function.

FIG. 6: Evolution of the H-function 6a, and in logarithmic scale 6b.

Next, we present the evolution of the H-function (Figure 6), comparing the two initial velocity distributions: the random cube (without norm one) and the random unit sphere (with norm one).Table III presents the values of the relaxation time for each distribution.

We observed that the system with the random unit sphere condition reaches equilibrium faster than the one with the random cube, for the case of the evolution of H and how the system follows the H-theorem, in the decreasing of the H-fuction. This difference arises from the distinct initial conditions, which is reflected in the relaxation time, τ_0. The relaxation time is lower for the random unit sphere condition compared to the random cube one.

V. CONCLUSION

We have studied the evolution of a non-equilibrium system towards equilibrium using kinetic theory, focusing on homogeneous systems, without external forces and on the approximation of the relaxation time. The results show a relationship between theoretical predictions and numerical simulations. However, slight discrepancies are observed because we consider a system of finite size, while the theoretical approximations work with an infinite system.

One of the conclusions we came to is that the velocity distribution at equilibrium is Maxwell-Boltzmann. We have verified that the moments of the velocities $\langle \mathbf{v}^p \rangle$ obtained in the simulation are in perfect agreement with the theory. Another result we get is that $\partial_t \langle v \rangle < 0$, $\partial_t \langle v^2 \rangle = 0$ and that $\partial_t \langle v^p \rangle > 0$ for $p > 2$. As p grows, the statistical fluctuations increase, due to larger values with larger p, increasing the amplitude.

We have found that for $\langle \mathbf{v}^4 \rangle$, systems with a smaller number of N particles reach equilibrium faster, moreover, the relation τ_0 and N follow a power law of exponent 0.04. The relationship between simulation and theory improves as the number of N particles increases. This demonstrates how the size of the system affects the relaxation dynamics.

The Boltzmann H function reached in the simulation coincides with the one calculated in theory, moreover, its evolution with time verifies the Boltzmann H Theorem, since it decreases with time until reaching its equilibrium value. Comparing the result for both initial conditions (unit sphere and cube) equilibrium is reached faster in the unit sphere, with shorter relaxation times.

It is observed that regardless of the initial conditions the systems reach equilibrium, demonstrating that it does not depend on it.

The discrepancy obtained in the relaxation time results for $\langle v^p \rangle$ and for H is due to the fact that in H a second approximation is made by considering $f^{(0)} \ln f \approx f^{(0)} \ln f^{(0)}$, causing the relaxation time of H to be smaller than that of $\langle v^p \rangle$.

APPENDIX A: NONLINEAR LEAST SQUARES FITTING

The data analysis was performed using the nonlinear least squares fitting method[4]. This approach employs the Levenberg–Marquardt algorithm, an iterative technique that interpolates between the Gauss–Newton method and gradient descent. Its objective is to minimize the chi-squared function, χ^2, thereby yielding the best-fit parameters for the model.

1. Methodology

Consider a dataset of N measurements (x_i, y_i), with associated uncertainties σ_i, and a model function $f(x;\theta)$ depending on parameters $\theta = (\theta_1, \ldots, \theta_m)$. The optimal parameters are those that minimize the following chi-squared function:

$$\chi^2(\theta) = \sum_{i=1}^{N} \left(\frac{y_i - f(x_i;\theta)}{\sigma_i} \right)^2, \tag{31}$$

as described in Ref.[5].

2. Implementation

The fitting procedure was implemented as follows:

1. The model function $f(x;\theta)$ was defined according to the physical description of the system.
2. The experimental data (x_i, y_i) were supplied to the curve fit, together with the model function.
3. The function returned the best-fit parameters θ_{opt} and the covariance matrix Σ.

3. Error Estimation

The covariance matrix Σ provides an estimate of the uncertainties in the fitted parameters. Its diagonal elements Σ_{jj} represent the variances of the parameters θ_j. The standard deviation (i.e., the statistical error) of a parameter θ_j is given by:

$$\sigma_{\theta_j} = \sqrt{\Sigma_{jj}}. \tag{32}$$

4. Fit Quality

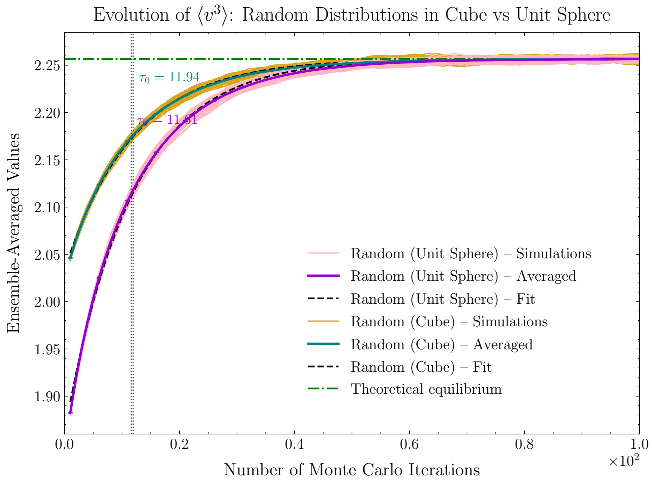
To assess the quality of the fits, we computed the coefficient of determination, R^2, defined as:

$$R^2 = 1 - \frac{\sum_i (y_i - f(x_i;\theta))^2}{\sum_i (y_i - \bar{y})^2}, \tag{33}$$

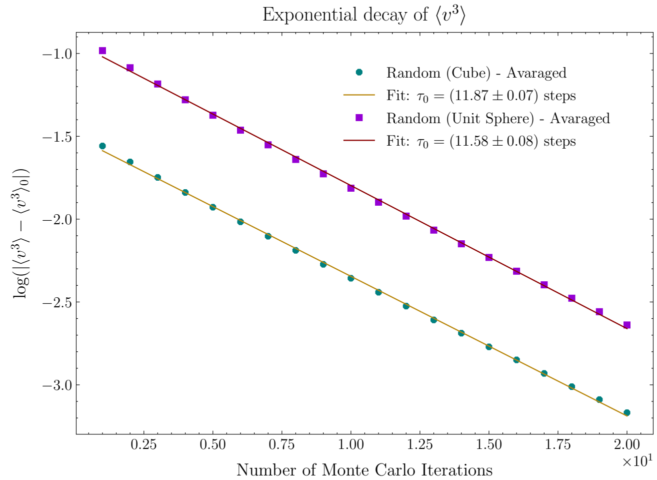
where $\bar{y}$ is the mean of the observed data. The R^2 metric quantifies the proportion of the variance in the data explained by the model, with values close to 1 indicating a good fit.

APPENDIX B: FIGURES OF THE EVOLUTION OF THE DISTRIBUTION OF THE VELOCITIES

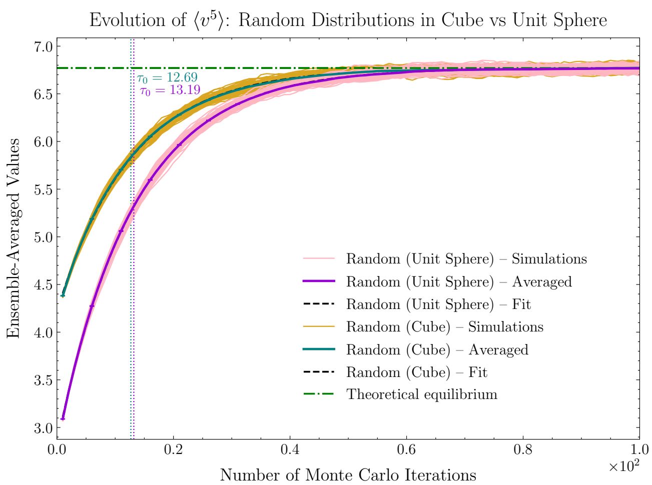
The next figures shown the evolution of the distribution of the velocities for $\langle \mathbf{v}^p \rangle \quad p \in \{3,5,6\}$ cases (Figure 7), and in the logarithmic scale for the $p = 1,4$ case (Figure 8).



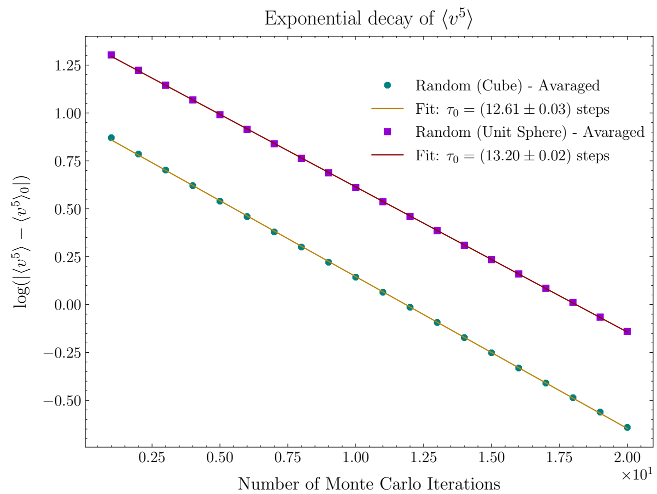
(a) Evolution of $\langle \mathbf{v}^3 \rangle$.



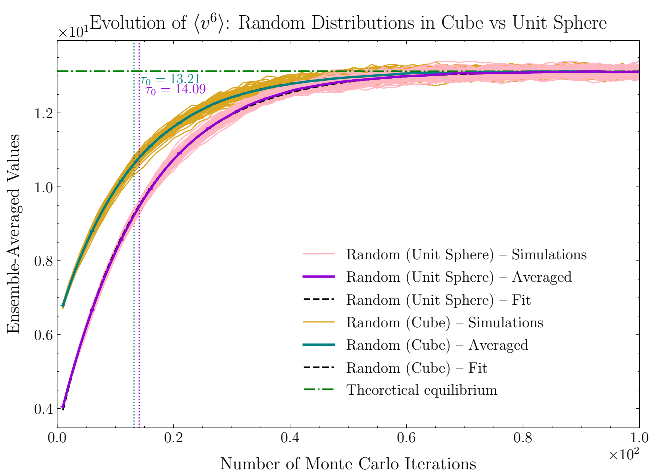
(b) Evolution of $\langle \mathbf{v}^3 \rangle$ in logarithmic scale.



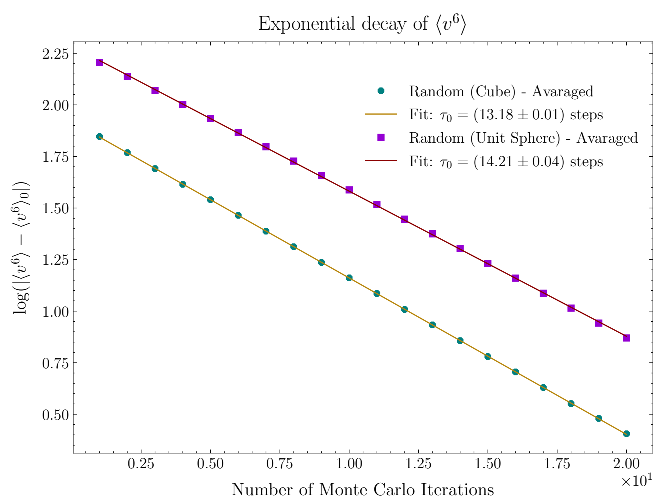
(c) Evolution of $\langle \mathbf{v}^5 \rangle$.



(d) Evolution of $\langle \mathbf{v}^5 \rangle$ in logarithmic scale.

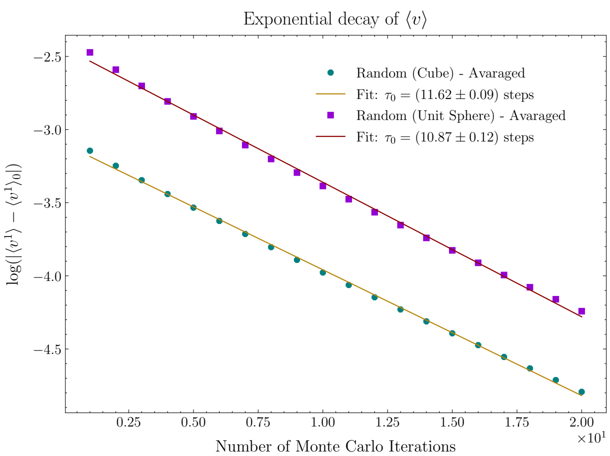


(e) Figure of the distributions for $\langle \mathbf{v}^6 \rangle$.

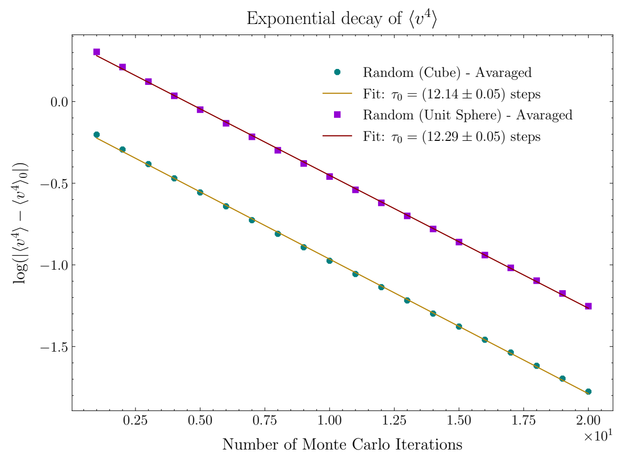


(f) Figure in logarithmic scale for $\langle \mathbf{v}^6 \rangle$.

FIG. 7: The first column of figures is for the distribution, while the other one is for the distribution in logarithmic scale. The Figures 7a and 7b correspond to the evolution of $\langle \mathbf{v}^3 \rangle$ with the number of Monte Carlo iterations; the Figures 7c and 7d for the $\langle \mathbf{v}^5 \rangle$ distribution of the velocities, and the Figures 7e and Figures 7f for thr $\langle \mathbf{v}^6 \rangle$ case.



(a) Figure in logarithmic scale for $\langle \mathbf{v} \rangle$.



(b) Figure in logarithmic scale for $\langle \mathbf{v}^4 \rangle$.

FIG. 8: Representation of the evolution of $\langle \mathbf{v} \rangle$ for Figure 8a and $\langle \mathbf{v}^4 \rangle$ for Figure 8b in logarithmic scale.

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